Dataset Cards

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach a card as an evidence-linked account.

Replace repeated audit tasks on the main route with current card-writing tasks while preserving saved earlier responses.

Keep one habit: Assess supported facts, explicit unknowns, proposed versus completed actions and bounded reuse conclusions.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Document a dataset’s source purpose row unit and version

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Use the three-session route to build a card from complete supplied fictional facts, distinguish unknown fields, and version a condition/count update. Earlier trip-audit material remains review. A card records **evidence** and questions; it does not certify quality, permission or model performance.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Retain original source and group IDs, exact assignments, intended prediction claim, label version and training-only preprocessing fit. A source record can document these without claiming that evaluation has happened.

Printed lesson and workbook

Supplied six-plant photo manifest, blank card headings and update record

Optional Python 3 editor; no packages required

Card headings: purpose source row/group counts conditions labels transformations splits limits version owner unresolved questions

Opening retrieval

Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

Look for: Choose a split unit that matches the intended **prediction** claim

Reference example, not the only acceptable wording: **Training** fits the model; validation compares development choices; test evaluates after choices are fixed.

Repeatedly selecting by the test score uses that set for development, so it is no longer an untouched final evaluation.

Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

Look for: Write an observable label rule with boundaries and a review option

Reference example, not the only acceptable wording: Touching is blocked. An object entirely outside is clear only when all objects are outside and the entire lane is visible.

A partly hidden lane requires review because clear cannot be established from the available description.

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

Look for: Identify a task from the meaning of its target

Reference example, not the only acceptable wording: Route type is classification because 1 and 2 encode categories. Whole-minute duration is regression because numerical differences measure time.

Exceeding 10 minutes is binary classification because the target is yes or no.

End goal

By the end, I can explain **dataset card**s using a traced example and a stated **limitation**.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Case exposure check: this source example appears in ai-4/5, ai-4/7, ai-4/9. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Leakage

Label Quality

Classification or Regression?

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Write five dataset-card fields for a leaf dataset and explain why '1000 images' alone is insufficient.

Concept to explain

A **dataset card** records where data came from, why it exists, its fields, collection **process**, known gaps, risks, and **prohibited use**s.

Term	Meaning
dataset card	structured documentation for a dataset
provenance	where data originated and changed
limitation	condition weakening valid use

Term	Meaning
prohibited use	use outside evidence or permission

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model source-linked entries and explicit unknowns

> Count photos and plant identities before interpreting the dataset.

Complete current worked card

Card field	Evidence-linked Photo practice v1 entry
Purpose / owner	Teacher-maintained fictional classroom card-writing and toy visible-spot practice
Source / version	Teacher-created fictional fixture, Photo practice v1; detailed creation date not supplied
Row / group / counts	One photo row; related views share plant ID;18photos/6plants/3views each

Card field	Evidence-linked Photo practice v1 entry
Conditions	One tablet; indoor fluorescent light; other conditions not documented
Labels	Vocabulary visible spot/no visible spot/review; per-photo labels and detailed annotation history not supplied
Changes / quality	No corrections or transformations supplied; completeness/measurement accuracy not established
Splits	Not assigned. A proposed unseen-plant evaluation must keep all views of a plant together; no evaluation has occurred
Intended / excluded uses	Classroom documentation practice; no demonstrated outdoor use or disease diagnosis
Open questions	Exact label rule/records, source detail, permission for any proposed external reuse, quality checks, collection dates and assignment record

Pictured facts and exact v2 update

Build a **dataset card** that reveals what the photo count hides.

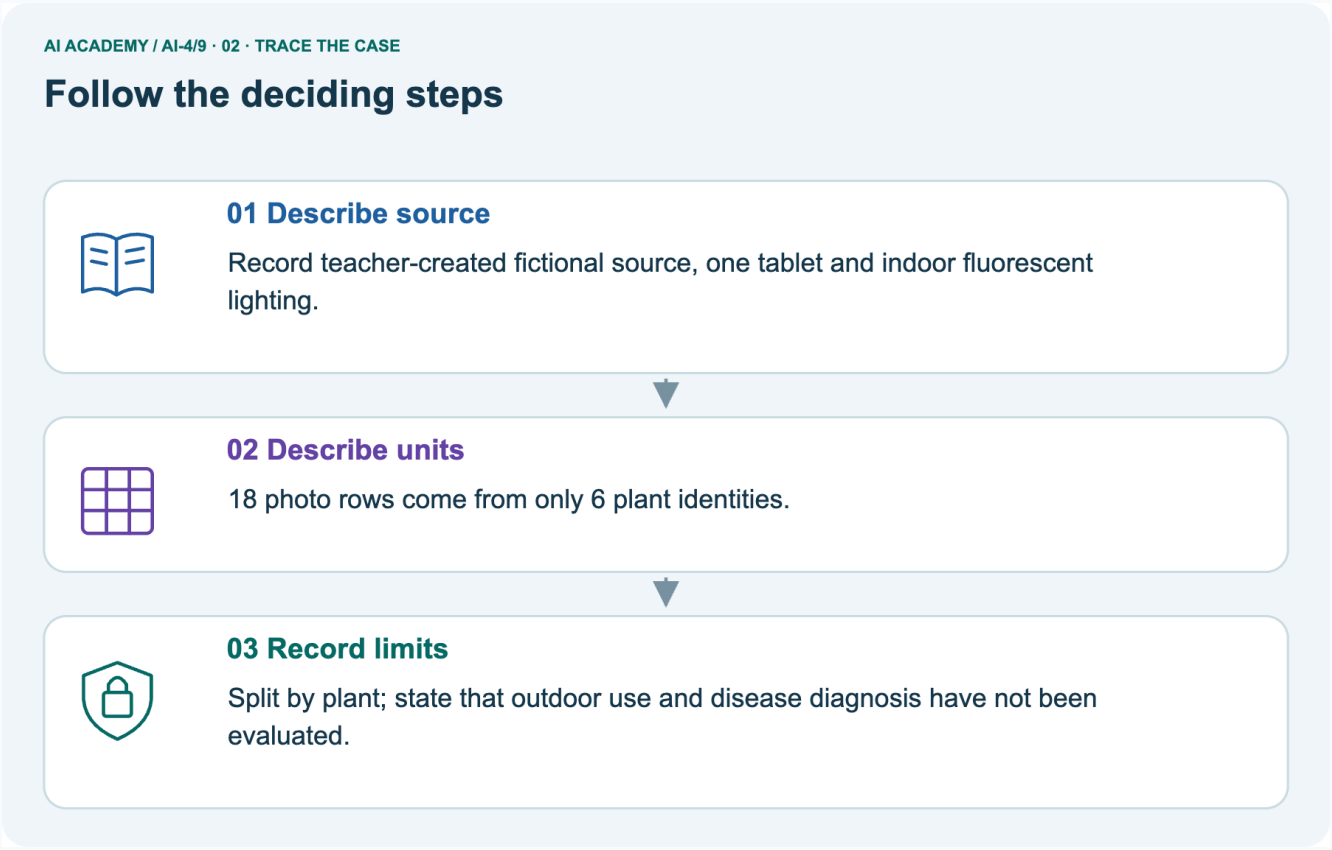
Teacher-created dataset: 18 photos of 6 fictional plants, 3 views each, one tablet, indoor fluorescent light. Label rule is visible spot/no visible spot/review.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Photo practice v2 adds one outdoor view of each of the same P1–P6 plants: new IDs P1d,P2d,P3d,P4d,P5d,P6d. The updated card records24 photos,6plant identities,18 indoor and6 outdoor views. Outdoor lighting details, device for the added views and per-photo labels are not supplied; do not invent them. A condition was added, but no additional source plants or accuracy result were created.

Preserve v1 and record this explicit change in v2.

Trace source, count and documented limits



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Describe source	Record teacher-created fictional source, one tablet and indoor fluorescent lighting.
2. Describe units	18 photo rows come from only 6 plant identities.
3. Record limits	Split by plant; state that outdoor use and disease diagnosis have not been evaluated.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Split by plant; state that outdoor use and disease diagnosis have not been evaluated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

A **dataset card** documents **evidence** and gaps; writing one does not fix biased sampling or inaccurate labels.

Predict the update and source-identity distinction

Change: The dataset adds 6 outdoor photos of the same plants.

Prediction: Photo count becomes 24, plant count remains 6; lighting coverage changes.

Reason: Additional views broaden conditions without creating new independent plant identities.

▼ Review · Earlier trip-audit card

A dataset can look like a neat table and still contain missing outcomes, repeated records or impossible values. Before **fitting** a model, inspect what one row represents and document where the table came from.

A **dataset card** is a short record of its purpose, contents, collection, changes and limits. It helps another person decide whether the data suits a proposed use.

Our source is a teacher-authored fictional export called Robot practice v1, created for this lesson. Each row is intended to describe one trip, with one unique trip ID and elapsed duration in minutes.

Durations may be zero or positive; None means unknown. All trips are described as indoor practice trips. No real robot measurements or personal records were collected, and there is no **evidence** about outdoor performance.

Keep this raw version unchanged. At this stage a flag means investigate, not automatically delete or replace. A repeated ID might be a copied row or a problem with the supposed one-row-per-trip definition.

We need **evidence** before deciding what happened.

A set stores each distinct ID once. The **output** is 5 4 2 1: five rows, four distinct IDs, two rows with unknown duration and one row with a negative duration. There is one extra occurrence beyond the four distinct IDs.

This is not a count of all rows in repeated-ID groups: both B rows belong to that group.

The code assumes every row is a pair and each duration is a number or None. It is a small audit, not full **schema** validation. It does not check text values, units, source truth or whether IDs refer to real unique events.

A successful run does not certify data quality.

Term	Meaning
Provenance	Where data came from and how it was created
Schema	The stated meaning types units and rules for fields
Missing value	An unknown or unavailable value rather than a measured zero

Term	Meaning
Dataset card	A concise record of dataset contents creation intended use and limitations

For Robot practice v1, record the purpose as practising inspection, the source as teacher-authored fictional data, and the row meaning as an intended single indoor trip.

Record the five-row count, the ID field, minutes as the duration unit and None as unknown. List the duplicate-ID and invalid-duration findings.

State that no cleaning has occurred, no train/test split is assigned, and no model has been evaluated.

Add intended use, excluded uses, known gaps, version and a contact or owner. Here the lesson teacher maintains the practice copy; its allowed classroom purpose is inspection practice.

Do not describe it as validated navigation data.

For a real external dataset, also verify its actual permission or licence and collection context; an absent field in the documentation is an unanswered question, not permission to invent a value.

> **What fraction of the five raw rows has unknown duration? Can you subtract two missing rows, one negative row and one extra ID occurrence from five to count distinct usable trips? Explain using the B rows.**

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Complete a **dataset card** and identify two uses the data cannot support.

MODEL IDEA A **dataset card** records where data came from, why it exists, its fields, collection **process**, known gaps, risks, and **prohibited use**s.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated **limitation**.

UNDER THE HOOD Source: student-collected classroom plant photos with permission. Purpose: teach a low-stakes healthy-versus-spotted classifier.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Source: student-collected classroom plant photos with permission.

STEP 2 Purpose: teach a low-stakes healthy-versus-spotted classifier.

STEP 3 Gap: few evening-light photos and no outdoor disease diagnosis **evidence**.

STEP 4 Prohibit medical, agricultural, or pesticide decisions; require teacher review.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

A **dataset card** lists fields but omits source and intended use.

Add information needed to assess suitability.

Exact explanation to model

Provenance, permissions, collection method/time/conditions, label definitions, coverage, exclusions, missingness, use and **limitation**s. Fields alone are insufficient.

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

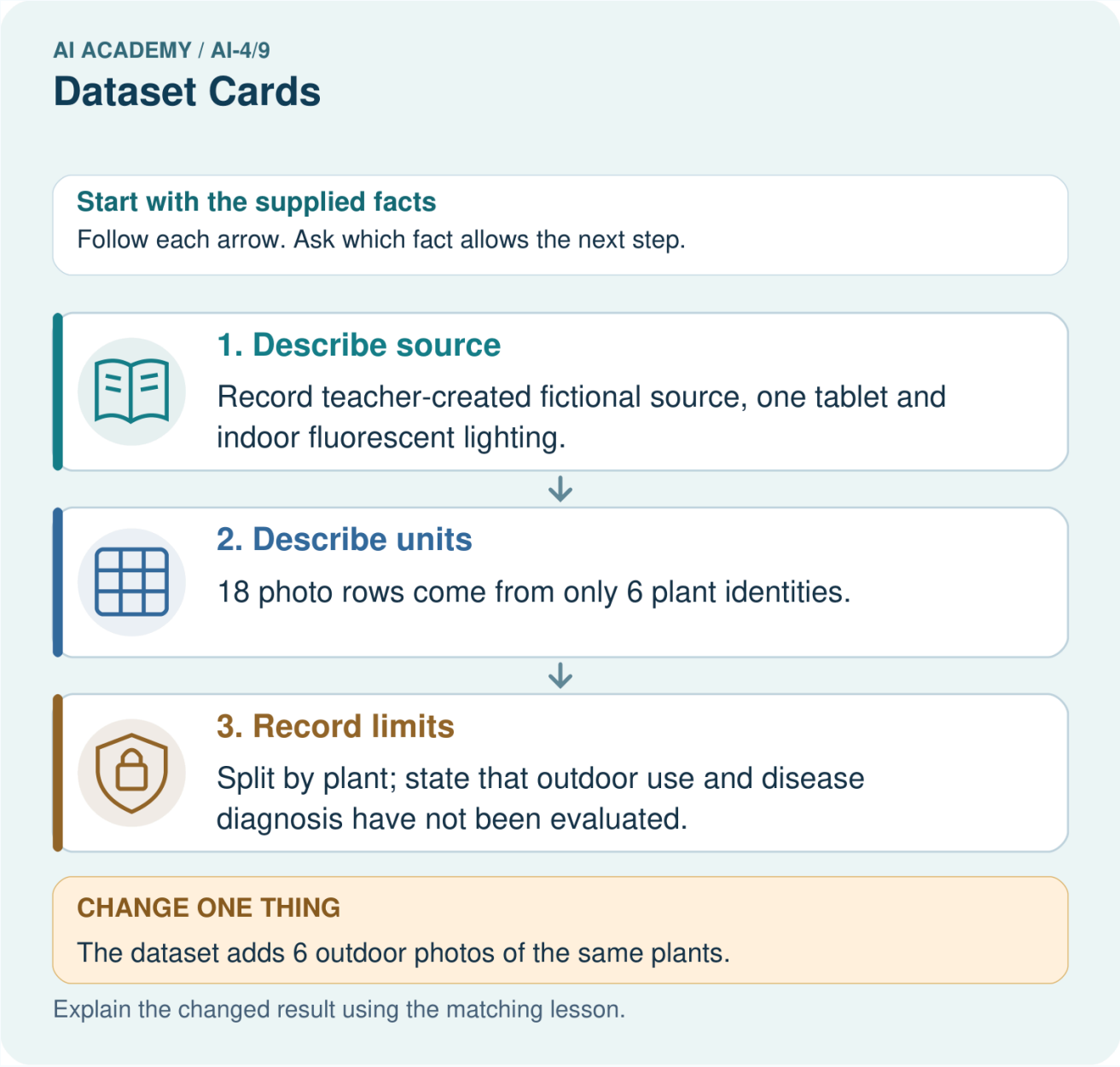
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Dataset Cards

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide card writing before the independent task

> Require an evidence source or an explicit unknown for every entry.

Current completion task and key

From Photo practice v1's complete manifest, fill purpose/source/version, photo and plant counts, known conditions, label status and split status. Mark one field not supplied and one use with no supporting evaluation. Then update the card for v2's P1d–P6d outdoor additions, preserving the earlier version.

v1 teacher-created fictional classroom fixture;18photos/6plants/3views each,one tablet,indoor fluorescent. Vocabulary given, per-photo labels/history not supplied. No split assigned or model evaluated. Unknown date/external-reuse permission must not be invented. No demonstrated outdoor use/disease diagnosis.

v2 adds6outdoor views of same6plants:24photos/6plants,18indoor/6outdoor; preserve v1 and note unknown added-view device/lighting details.

Current misconception check

A dataset card has every heading filled, but each unknown field was guessed. Does completeness establish that its claims are correct or that a model is ready? Name the repair.

Completeness does not validate guesses. Mark unknowns, ask for evidence, preserve versions and perform the missing checks.

Pictured update reasoning

What two counts should the revised card show? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

24 photos and 6 plants, with view relationships, source and indoor/outdoor conditions documented. Changed-case check: Photo count becomes 24, plant count remains 6; lighting coverage changes.

Additional views broaden conditions without creating new independent plant identities.

Changed-case reasoning: Photo count becomes 24, plant count remains 6; lighting coverage changes. Additional views broaden conditions without creating new independent plant identities.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: A **dataset card** lists fields but omits source and intended use.

Write the question to explain: Add information needed to assess suitability.

Reveal after **prediction**s: **Provenance**, permissions, collection method/time/conditions, label definitions, coverage, exclusions, missingness, use and **limitation**s. Fields alone are insufficient.

Use the earlier transfer question as additional practice. Reserve the selected20-photo five-plant documentation task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: 2 of 5 raw rows have unknown duration, or 40 percent. No: the extra B occurrence is already among the missing rows, so subtracting all three flag counts double-counts it.

A distinct usable-trip count requires a stated resolution policy and source review. Exit examples include duplicate IDs, impossible values, units, **provenance** and coverage.

A defensible statement is that this five-row fictional export has documented quality flags and is suitable for inspection practice, with no model-readiness claim.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve regression target meaning and why later outcomes are separate from prediction inputs. Establish that this week inspects data before fitting.
7 to 18 minutes	Read the source description and schema. Count physical rows and distinct trip IDs separately. Ask whether a repeated ID proves the reason for duplication.
18 to 32 minutes	Trace the code with row cards. Explain set, None and negative checks. Complete W1 and W2 with named denominators.
32 to 45 minutes	Use the guided question to show overlap: place both B cards in missing and repeated-ID groups. Do not subtract all flag counts independently.

Stage	Teacher action
Session 2 start to 8 minutes	Retrieve raw version versus corrected version. Ask what evidence could justify changing a recorded value.
8 to 20 minutes	Read the card example and complete W3. Distinguish a documented fact, an unknown and an assumption.
20 to 35 minutes	Collect W4 independently. Require counts, overlap and a concise card with source and limitations before offering hints.
35 to 45 minutes	Discuss W5 and the exit claim. Use the fresh retry after support and record the support given.

Reduce help deliberately

Completion task: Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

Teacher check: The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The card records **evidence** and gaps; its claims still need checking. Documentation does not prove suitability.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete card-writing and version review

> Keep supplied evidence separate from unknowns and future actions.

Current fixture and answer guidance

v1 teacher-created fictional classroom fixture;18photos/6plants/3views each,one tablet,indoor fluorescent. Vocabulary given, per-photo labels/history not supplied. No split assigned or model evaluated. Unknown date/external-reuse permission must not be invented. No demonstrated outdoor use/disease diagnosis.

v2 adds6outdoor views of same6plants:24photos/6plants,18indoor/6outdoor; preserve v1 and note unknown added-view device/lighting details. Independent key:20photos/5plants/4views,one phone/indoor; growth labels absent, creator/date/permission not given. Mark unknown, propose target/rule and grouped split; do not invent completed labels, assignments or performance.

Predict, document and inspect

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. **Dataset Card** Review: inspect two incomplete cards, find missing **provenance** or risks, and write a defensible replacement.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the evidence ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose current card-writing errors

Card misconceptions: row count equals independent sources—report both; vocabulary means labels exist—state missing assignments; a filled heading proves truth—require evidence; proposed grouping is completed split—mark status; outdoor views add plants—preserve plant IDs; absent permission can be guessed—record a question and resolve it for the intended use.

Choose support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected20-photo five-plant documentation task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR A file name and list of columns are enough documentation for responsible reuse.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A complete card certifies data quality: The card records **evidence** and gaps; its claims still need checking. Documentation does not prove suitability.

Missing means zero: None is unknown; zero is a specific allowed value in this exercise. Substitution without **evidence** changes the meaning.

Flag counts can always be added: The same row can have multiple flags. Count rows or unique events under an explicit policy and retain the raw denominator.

Match the support to the first error
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MAIN LESSON · SECTION 5

Assess the selected fresh dataset card

> The current workbook tests documentation reasoning rather than copied trip-audit counts.

Selected task and checked answer

Selected case: Write key dataset-card entries for20 photos: five plants, four views each, one phone, one indoor setting. Growth labels have not been supplied.

Record collection source,20 images and five plant groups, view/device/setting coverage, missing label definitions, intended use and **limitation**s. Specify a plant-group split and obtain a defensible label rule.

Twenty images are not20 independent plants; no accuracy claim is supported without labels and evaluation.

Current W1–W3/W5 guidance

Current W1–W2 require v1 facts and explicit unknowns; v2 is24photos/6plants,18indoor/6outdoor. W3 rejects guessed completeness. W5 identifies absent growth labels/outdoor evidence and unresolved intended-reuse questions, with no suitability guarantee. v1 teacher-created fictional classroom fixture;18photos/6plants/3views each,one tablet,indoor fluorescent. Vocabulary given, per-photo labels/history not supplied.

No split assigned or model evaluated. Unknown date/external-reuse permission must not be invented. No demonstrated outdoor use/disease diagnosis. v2 adds6outdoor views of same6plants:24photos/6plants,18indoor/6outdoor; preserve v1 and note unknown added-view device/lighting details.

Judge evidence and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

▼ Review · Earlier audit questions and keys

W1 Define the row and source

For Robot practice v1, state what one row is intended to represent, the source, the unit and the meaning of None.

One row is intended to represent one indoor practice trip. The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

W2 Interpret the audit counts

Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

W3 Reject an unsupported correction

A learner changes None to 0 and -2 to 2, then reports a clean dataset. Explain why this is not a justified repair and what record should be kept.

Those edits invent outcomes without source **evidence**. Preserve the raw version, flag the affected rows and investigate.

Record any supported correction with old value, new value, reason and version; unresolved values remain uncertain.

W4 Inspect and document independently

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

There are 6 rows, 5 distinct IDs, 2 missing durations, 1 negative duration and 1 extra ID occurrence. Both Q rows are missing, so the duplicate and missing flags overlap. Zero for T is allowed by the stated **schema**.

The card should identify the supplied fictional indoor export, one intended row per trip, minutes, all findings, raw unchanged version and no **evidence** for real or outdoor navigation use.

W5 State an evidence gap

Someone duplicates each row ten times and claims the export now covers more kinds of trips. Explain what changed and what did not.

The row count increased but no new trip conditions or independent **observation**s were added. Coverage remains limited to the same supplied fictional indoor cases; the larger count does not establish broader usefulness.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Provenance, permissions, collection method/time/conditions, label definitions, coverage, exclusions, missingness, use and **limitation**s. Fields alone are insufficient.

Additional transfer practice question

A **dataset card** lists 500 images but omits collection conditions and intended use. Add two essential entries and explain their value.

Reasoned practice answer

Document where and how images were collected and the intended or excluded uses. These help readers judge suitability, coverage and limits beyond row count.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The number of rows alone adequately describes a dataset.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Include **provenance**/permission, collection conditions and dates, leaf identity groups, operational label definitions and class/condition coverage, plus missingness, intended use and exclusions.

A large count can consist of duplicates or narrow conditions.

Explanations for the additional workbook examples

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Publish the Plant Classifier **dataset card** before model experiments begin; update it when data changes.

Keep the decision, **input**, result, explanation and remaining **limitation** in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Offer blank card headings and the exact source brief; prompt learners to label entries given, unknown or proposed. Extension: require a revision record with old/new counts, unchanged plant IDs and unresolved metadata.

Selected independent key:20photos/5plants/4views each,one phone/indoor,growth labels absent. Source creator/date/permission and other unsupplied fields remain unknown; require a proposed target/rule and plant grouping without claiming completed evaluation. Fresh retry:12photos/4objects/3views,one indoor device,labels unknown; require row/group counts, evidence-linked fields, unknowns and an explicitly proposed split.

Delayed return

Suggested return points in this level: ai-4/10, ai-4/12, ai-4/15. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 10: Baselines First. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week10: Baselines First. Carry forward the versioned card, source groups, label contract and intended use so every baseline result names its data and evaluation context. Research reference: https://arxiv.org/abs/1803.09010 (Datasheets for Datasets).

▼ **Earlier wording, alternatives and picture transcripts**

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Inspect row meaning **missing value**s and repeated records

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

Reasoned practice answer: The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

Check every learner before discussion: Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The card records **evidence** and gaps; its claims still need checking. Documentation does not prove suitability.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The dataset adds 6 outdoor photos of the same plants. Require a **prediction** and a reason before comparing. Expected result: Photo count becomes 24, plant count remains 6; lighting coverage changes.

Reason: Additional views broaden conditions without creating new independent plant identities.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and **limitation**s. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

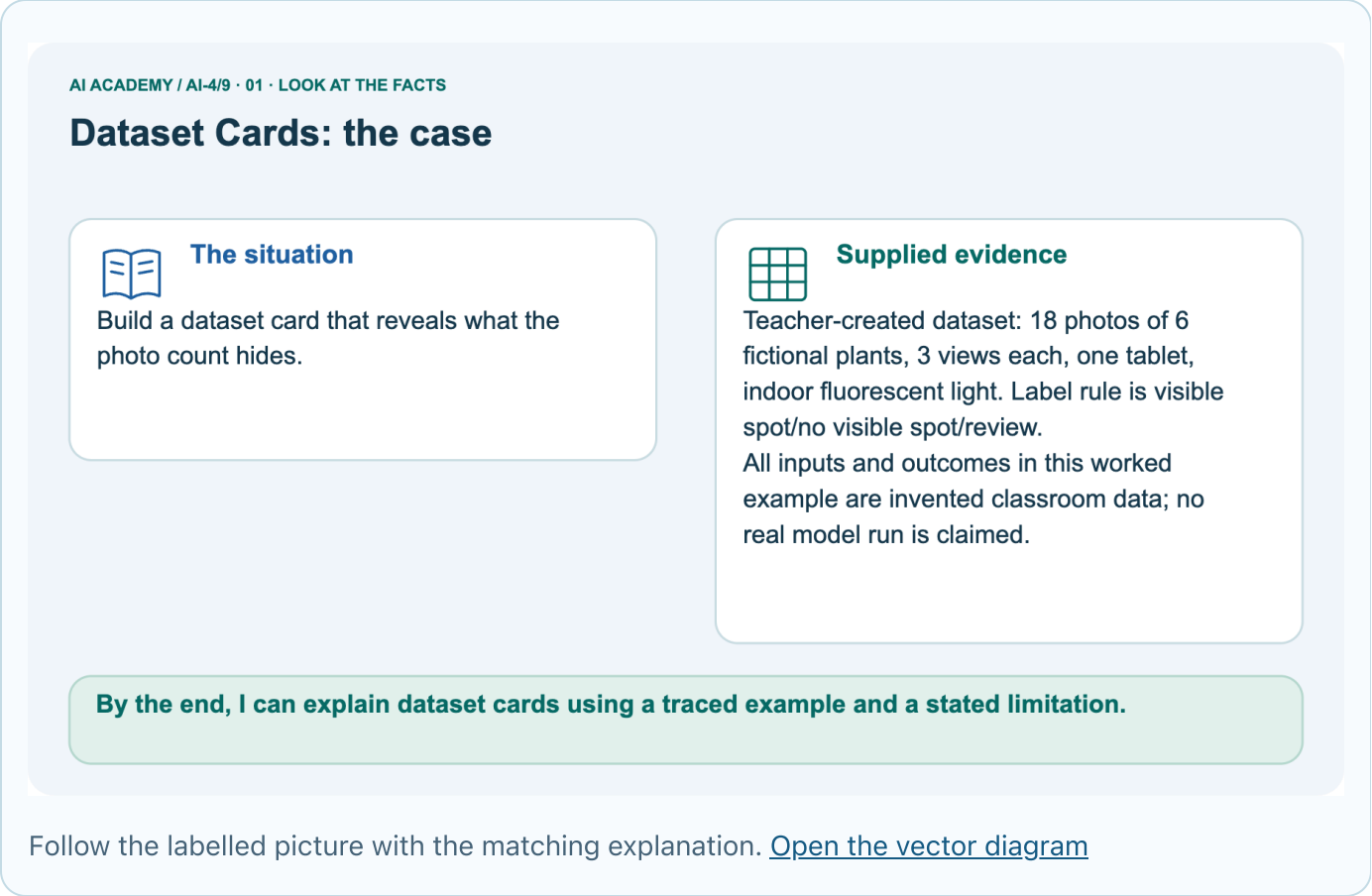
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Dataset Cards: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

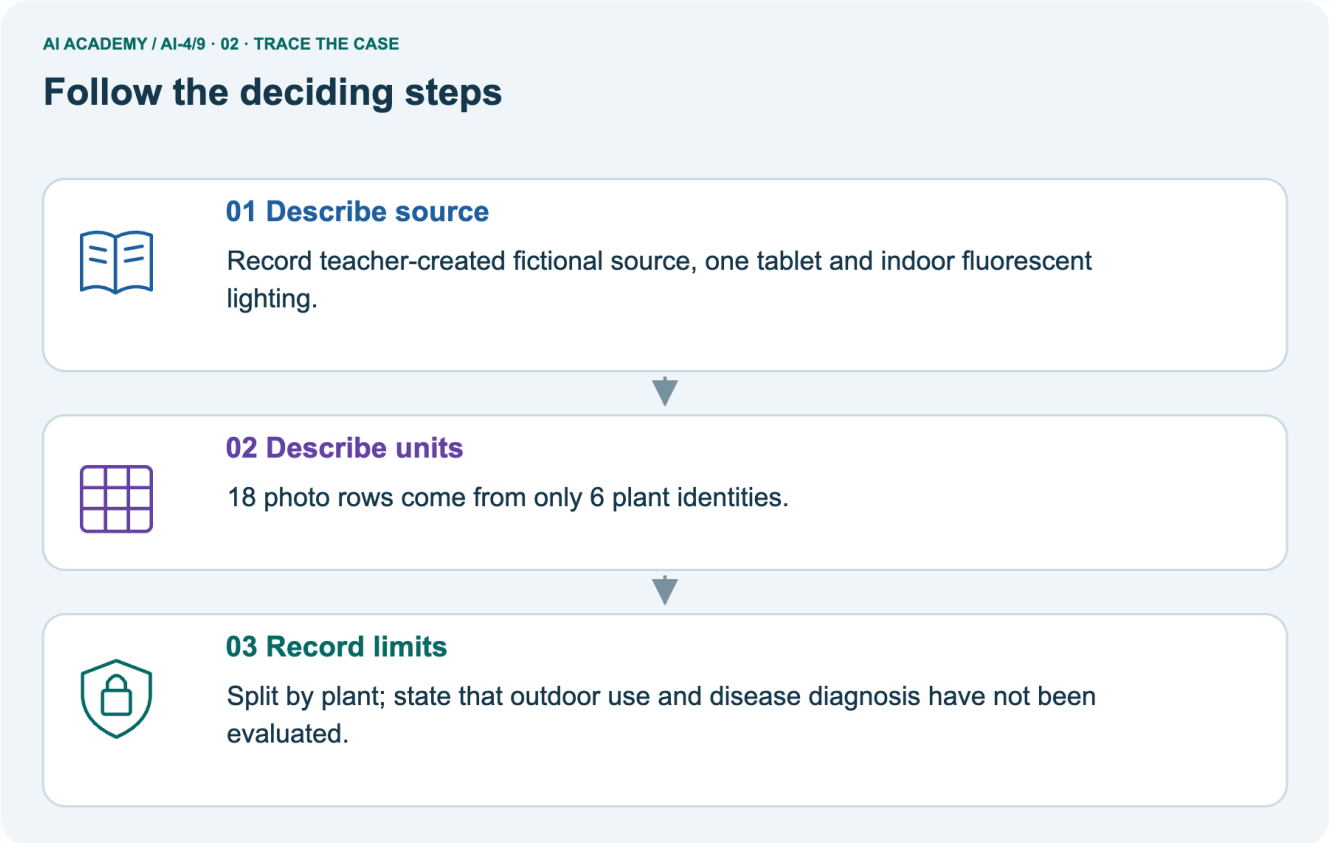
Build a **dataset card** that reveals what the photo count hides. Supplied facts: Teacher-created dataset: 18 photos of 6 fictional plants, 3 views each, one tablet, indoor fluorescent light.

Label rule is visible spot/no visible spot/review. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



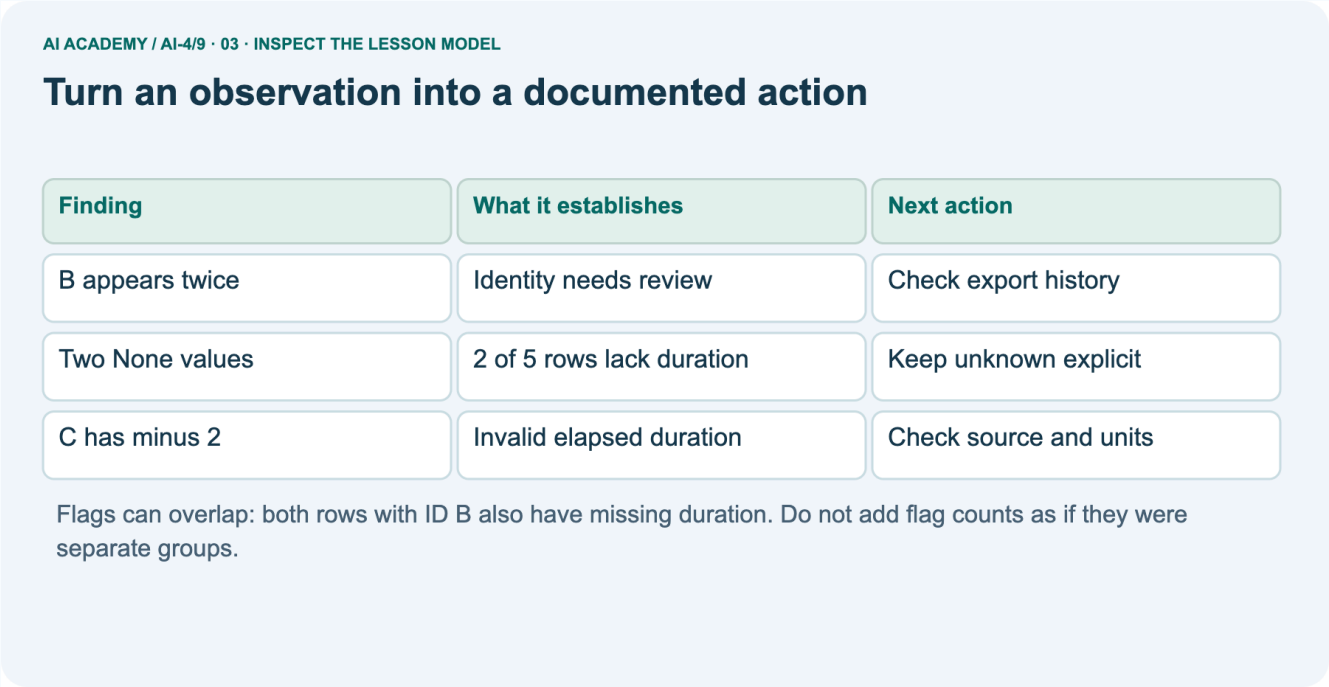
Follow the labelled picture with the matching explanation. Open the vector diagram

Describe source: Record teacher-created fictional source, one tablet and indoor fluorescent lighting. → Describe units: 18 photo rows come from only 6 plant identities. → Record limits: Split by plant; state that outdoor use and disease diagnosis have not been evaluated.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Turn an observation into a documented action



Follow the labelled picture with the matching explanation. Open the vector diagram

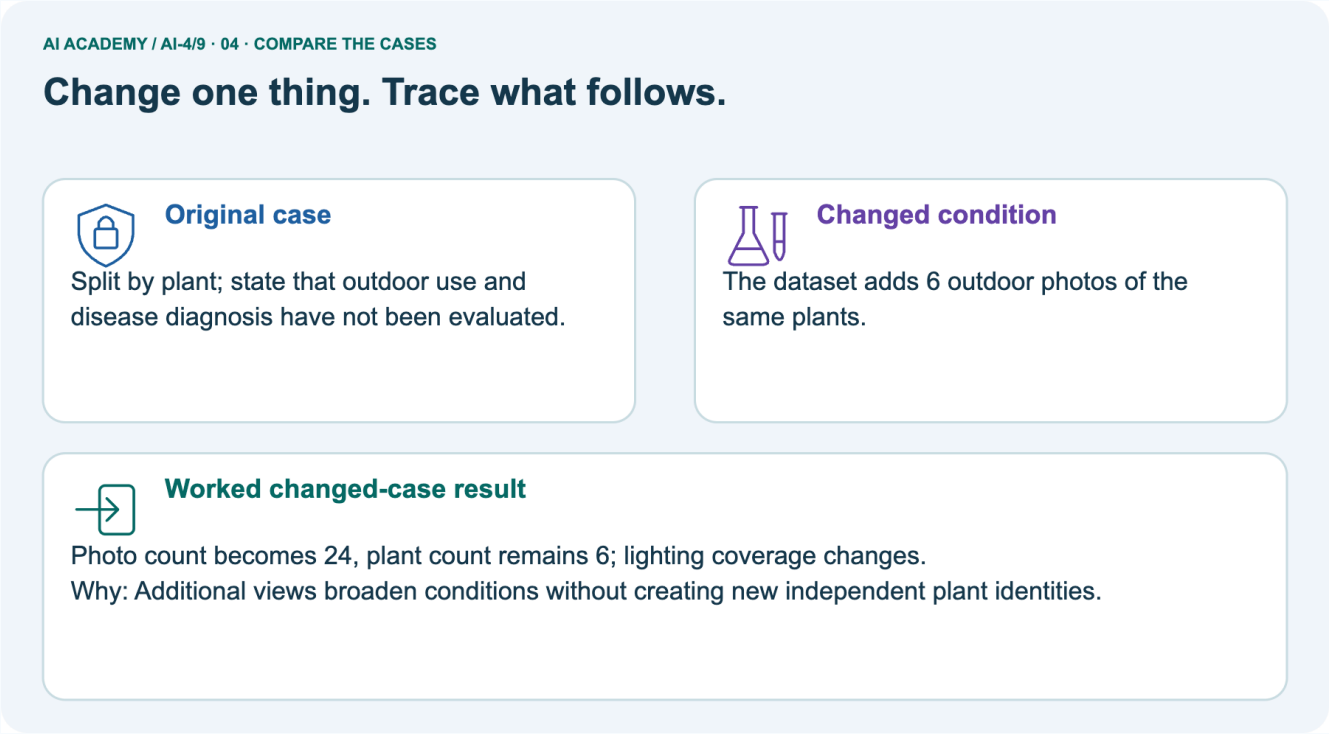
Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Split by plant; state that outdoor use and disease diagnosis have not been evaluated. Changed condition: The dataset adds 6 outdoor photos of the same plants.

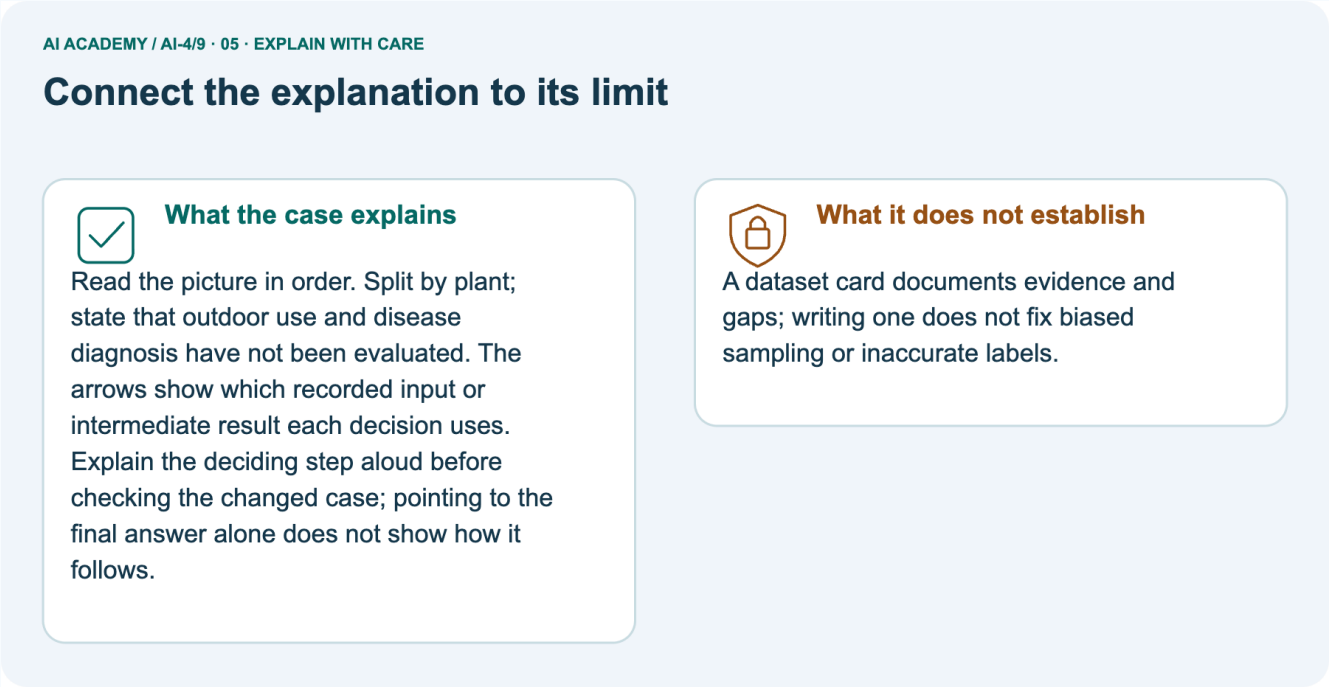
Expected result: Photo count becomes 24, plant count remains 6; lighting coverage changes. Reason: Additional views broaden conditions without creating new independent plant identities.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Additional views broaden conditions without creating new independent plant identities.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Split by plant; state that outdoor use and disease diagnosis have not been evaluated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

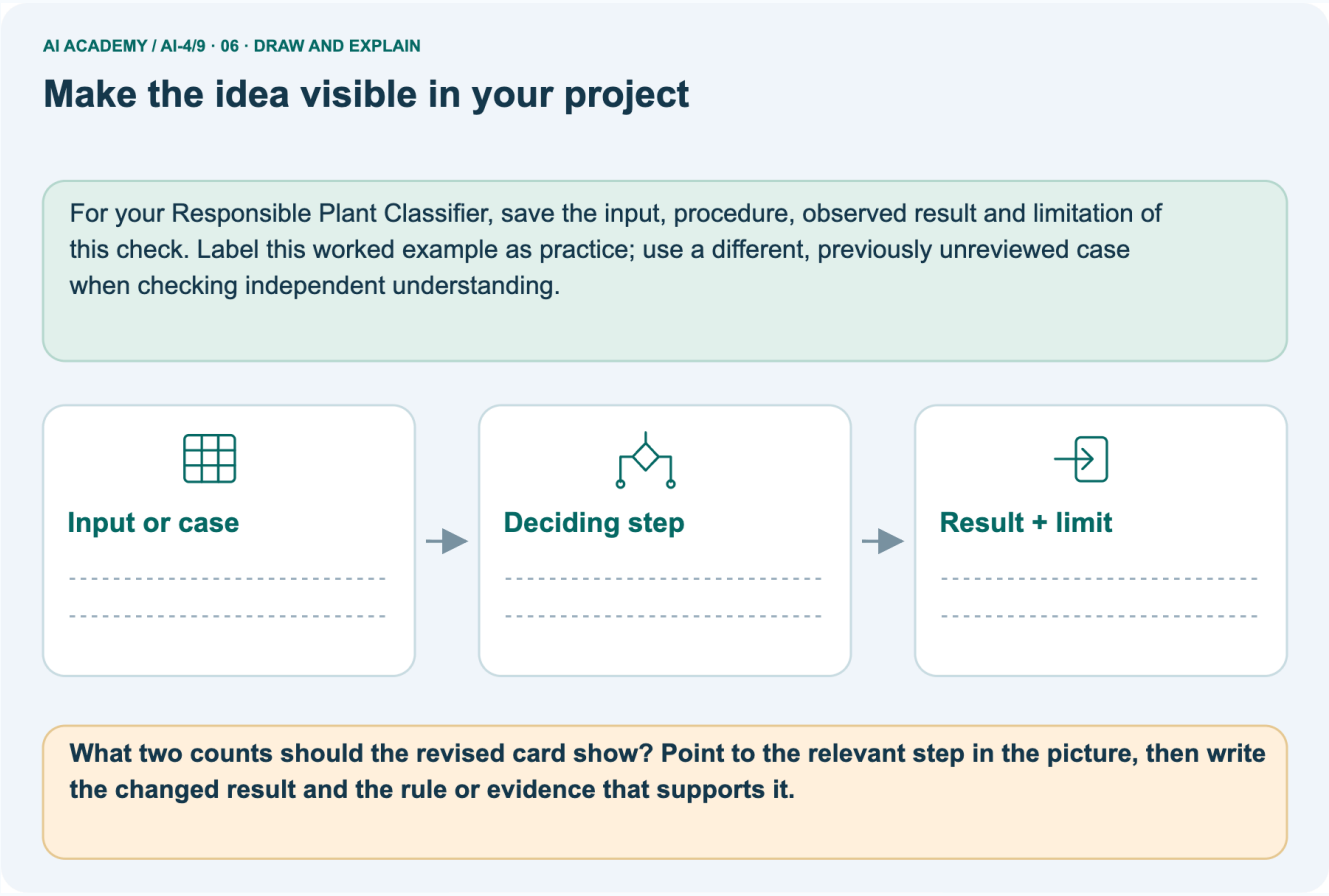
Limit: A **dataset card** documents **evidence** and gaps; writing one does not fix biased sampling or inaccurate labels.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A **dataset card** documents **evidence** and gaps; writing one does not fix biased sampling or inaccurate labels.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What two counts should the revised card show? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: 24 photos and 6 plants, with view relationships, source and indoor/outdoor conditions documented. Changed-case check: Photo count becomes 24, plant count remains 6; lighting coverage changes.

Additional views broaden conditions without creating new independent plant identities.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Dataset Cards: the case

02. Follow the deciding steps

03. Turn an observation into a documented action

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.